

Usman Ahad

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EDUCATION

Lahore University of Management Sciences

Lahore, Punjab, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.64 / 4.00)

Aug. 2023 – Present

- **Honors:** Placed on the Dean's Honour List for consecutive academic years (2023-24, 2024-25, and 2025-26).
- **Graduate Coursework:** Advanced Topics in Machine Learning (CS 6304), AI on Edge Devices (AI 624), Deep Learning (AI 623), Internet of Things (CS 677).
- **Core Coursework:** Foundations of AI & ML, Advanced Topics in Machine Learning, Deep Vision Language Models, AI on Edge Devices

RESEARCH PROJECTS

VisConf: Quality-Aware Consensus for Best-of- N VLM Reasoning

PyTorch, Transformers, Qwen2.5-VL, InternVL3, Gemma 4

Jan. 2026 – Sep. 2026

- Developed a training-free Best-of- N framework combining candidate-calibrated Self-Certainty with visual-attention engagement and hyperparameter-free Rank-Weighted Consensus, improving VLM answer selection without external verifier or reward models.
- Evaluated 3 VLMs across MathVista, MMMU-Pro, and MMStar at $N \in \{8, 16, 32\}$; achieved the highest average accuracy at every model-budget setting and outperformed all baselines in **24 of 27** model-dataset-budget configurations.
- Reached **56.36%** mean accuracy, exceeding Self-Consistency by 0.98 points and the strongest baseline by 0.49 points (paired 95% CI: [0.32, 0.66]); delivered gains up to 12.18 points when correct rollouts were present but outnumbered.

Mind Your Own Business: Consent-Based Audio Privacy for Smart Glasses

PyTorch, Qwen3.5-2B, LoRA, DPO, RAG, LS-EEND, Speech Processing

Jan. 2026 – Sep. 2026

- Built a paced, semi-real-time Raspberry Pi 5 pipeline that diarizes and transcribes speech, tracks cumulative disclosures across rolling windows, anonymizes linked and linkable identifiers, resynthesizes speaker-independent audio, and cryptographically gates restoration on bystander consent.
- Distilled GPT-5.6 Luna anonymization into Qwen3.5-2B using 30,960 SFT examples and 6,241 DPO preference pairs; the Q4 edge variant retained 0.902 utility with 0.400 privacy leakage on 143 held-out transcripts.
- Reduced mean privacy leakage by **24.7%** versus the 2B base model (0.502 $\rightarrow$ 0.378) while raising utility by 2.5 points (0.875 $\rightarrow$ 0.900); stayed within 6.5% of GPT-5.6 Luna's leakage while exceeding its utility by 1.4 points.

FedCPR: Contrastive Learning w/ Latent Prototype Rejection

PyTorch, Federated Learning, Open-Set Recognition

Nov. 2025 – Feb. 2026

- A communication-efficient federated learning framework addressing open-set recognition (OSR) challenges where distributed edge models encounter unknown classes during deployment.
- Developed Latent Prototype Rejection (LPR) to dynamically perturb intermediate features rather than raw pixels for synthetic outlier generation under standard FedAvg aggregation; incorporated a curriculum learning schedule to progressively ramp perturbation difficulty and stabilize gradient updates across non-IID clients.
- Achieved 85.42% open-set AUROC and 90.96% closed-set accuracy on CIFAR-10 across 5 heterogeneous clients, significantly outperforming SOTA alternatives including PROSER (80.77%), FedPD (80.58%), and ARPL (73.41%).

WatchTower: Transformer-Based Network Anomaly Detection for Edge Devices

PyTorch, TFLite, Apache TVM, FedProx, PROSER, Raspberry Pi 5

Aug. 2025 – Dec. 2025

- Built a real-time open-set network intrusion detector for Raspberry Pi 5 using 28 packet-level features, a memory-token Transformer with gated cross-window state, PROSER latent-space outliers, and FedProx for heterogeneous router traffic.
- Achieved **93.66% accuracy** and **92.83% F1** on CIC-IDS2017, improving over the evaluated LSTM-AE baseline by 19.56 and 10.33 percentage points, respectively; validated live DoS detection across 3–8 connected devices.
- Compressed the FP32 model by **67.5%** (4.63 MB $\rightarrow$ 1.51 MB) through INT8 TFLite quantization, then ran 1,000-trial TVM auto-tuning for ARM Cortex-A76 to reach 3.82 ms single-core latency (260+ packets/sec), a $2.6\times$ speedup.

TEACHING EXPERIENCE

Teaching Assistant for AI on Edge Devices

Aug. 2026 – Present

Lahore University of Management Sciences

Lahore, Punjab, Pakistan

- Support a 15-week graduate course spanning performance measurement, quantization, hardware-aware compilation, TinyML, edge LLMs, and federated privacy across microcontroller, Raspberry Pi, Jetson, and Edge TPU targets.

Teaching Assistant for Deep Learning

Jan. 2026 – May 2026

Lahore University of Management Sciences

Lahore, Punjab, Pakistan

- Supported 100+ MS AI students through programming assignments, technical office hours, and evaluation across CNNs, sequence models, transformers, VAEs, GANs, and diffusion models.

Teaching Assistant for CS 200 (Introduction to Programming)

Jan. 2025 – May 2025

Lahore University of Management Sciences

Lahore, Punjab, Pakistan

- Mentored 200+ undergraduate students in C++ architecture and low-level memory management concepts (pointers, manual allocation).
- Debugged complex student codebases and evaluated algorithmic efficiency, reinforcing core systems programming paradigms.

TECHNICAL SKILLS

Languages: C/C++, Java, SQL, JavaScript, HTML/CSS, MATLAB

Frameworks & Libraries: PyTorch, Transformers, Hugging Face, ONNX Runtime, TFLite, Apache TVM, Scikit-learn, NumPy, pandas, Librosa

Speech & Audio AI: LS-EEND, Streaming ASR, Speaker Embeddings, Speech Separation, Neural TTS, Audio Anonymization

Developer Tools & Platforms: Git, Linux/Unix Shell, Docker, Jupyter Notebooks, Wireshark, Raspberry Pi 5, Edge Hardware

Theoretical Expertise: Open-Set Recognition, Federated Learning (FedProx/FedAvg), Preference Optimization (DPO/LoRA), Multimodal Evaluation (VLMs), Natural Language Processing, Neural Architecture Search, Hardware-Aware Compilation